

DDM-VLM: PSYCHOLOGY-INSPIRED DIFFUSION DECISION MODELING FOR VISION-LANGUAGE WAFER DEFECT RECOGNITION

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ABSTRACT

Wafer map defect recognition is challenging due to diverse spatial patterns and the frequent co-occurrence of multiple defect types. While vision-language models (VLMs) enable training-free recognition, a single response may be unreliable for ambiguous joint defect states. We introduce a sequential decision framework inspired by the diffusion decision model (DDM) in cognitive psychology, which treats VLM responses as noisy observations and progressively accumulates evidence over valid multi-hot defect labels. We further retrieve similar wafer maps as visual memory cues to provide complementary evidence and candidate conditioning. The decision process terminates when the evidence margin between the two leading labels exceeds a predefined boundary. Experiments demonstrate that our method consistently outperforms both zero-shot VLMs and conventional retrieval-augmented generation (RAG) baselines across multiple VLMs. The source code is publicly available at <https://github.com/your-repository>.

Index Terms— wafer map, diffusion decision model, vision-language model, sequential decision making, multi-label classification

1. INTRODUCTION

Wafer map defect recognition is important for semiconductor yield analysis, as spatial failure patterns reveal manufacturing abnormalities such as edge effects, contamination, scratches, and process variations. Unlike conventional image classification, wafer maps are low-resolution symbolic grids with limited texture, and multiple defect patterns may coexist. Mixed-defect recognition is therefore particularly challenging, as it requires predicting a complete multi-hot joint defect state rather than a single isolated class.

Wafer defect recognition has been widely studied using machine- and deep-learning methods [1], including CNN-based recognition and retrieval [2, 3], mixed-defect recognition [4, 5, 6, 7, 8, 9], and more recent contrastive, generative, few-shot, and transformer-based approaches [10, 11, 12, 13, 14]. Despite their effectiveness, these methods generally rely on task-specific training. Recent vision-

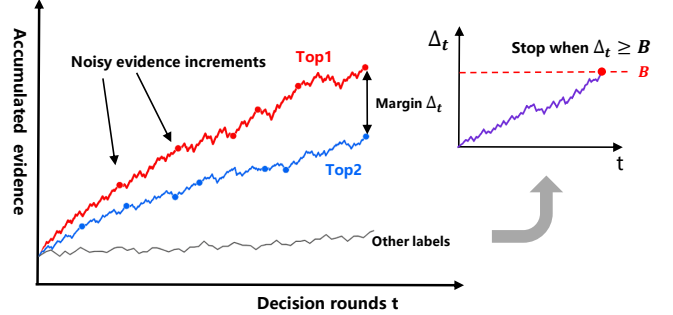


Fig. 1. Conceptual illustration of the DDM-inspired sequential decision process. Evidence for competing joint labels is accumulated across decision rounds, and inference stops when the top-two margin reaches the boundary.

language models (VLMs) provide a flexible, training-free alternative by reasoning over wafer images and textual defect descriptions. Cao et al. [15], for example, explored visual-textual prompting for zero-shot wafer classification. However, extending this paradigm to mixed-defect recognition is substantially more difficult: MixedWM38 contains 38 joint states formed by eight potentially co-occurring defects, whose visual differences can be subtle. Consequently, directly adopting a single VLM response can be unreliable and prompt-sensitive, especially when several defects coexist.

We address this problem from the perspective of the diffusion decision model (DDM), a classical sequential evidence-accumulation framework in cognitive psychology [16, 17, 18]. Rather than making a decision from a single noisy observation, DDM integrates evidence over time until sufficient separation between competing hypotheses is reached. This mechanism naturally fits ambiguous mixed-defect recognition: successive VLM responses can provide additional perceptual evidence, while visually similar labeled wafers serve as external visual-memory cues. As illustrated in Fig. 1, evidence for competing joint labels is accumulated across decision rounds until the margin between the leading candidates reaches a predefined boundary.

Based on this idea, we propose DDM-VLM, a psychology-inspired sequential decision framework for VLM-based wafer defect recognition. An external multi-alternative accumulator maintains evidence over valid joint labels. Across succes-

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sive rounds, VLM answer probabilities provide perceptual evidence, while retrieved-wafer similarities provide complementary memory evidence. These signals are sequentially integrated with diffusion noise, and inference terminates once the evidence margin between the two leading labels reaches a decision boundary. Thus, retrieval serves as visual-memory conditioning rather than the final decision mechanism, while DDM-style evidence accumulation determines how uncertain observations are integrated into a joint prediction.

Our contributions are threefold. First, we formulate VLM-based mixed wafer recognition as a psychology-inspired sequential decision problem rather than a one-shot prediction task. Second, we introduce a multi-alternative DDM mechanism that accumulates evidence over complete joint defect states and performs adaptive boundary-based termination. Third, we integrate VLM perceptual evidence with retrieval-based visual memory evidence and demonstrate consistent improvements over a retrieval-augmented single-response baseline across multiple VLM backbones.

2. METHOD

2.1. Preliminary: Diffusion Decision Model

The diffusion decision model (DDM) is a classical framework for sequential decision making under uncertainty [16, 17, 18]. In the standard two-alternative setting, a latent decision variable accumulates noisy evidence over time,

$$dz(t) = v dt + \sigma dW(t), \quad (1)$$

where v denotes the mean evidence drift, σ controls the stochastic diffusion, and $W(t)$ is a Wiener process. The drift determines the average direction and rate of evidence accumulation, whereas the diffusion term models trial-to-trial uncertainty in the observed evidence.

A decision is made when the accumulated evidence reaches a predefined boundary. Strong evidence produces a larger drift and typically reaches the boundary earlier, while ambiguous observations require longer accumulation. The boundary thus controls a speed-accuracy trade-off: a smaller boundary permits faster but less conservative decisions, whereas a larger one requires stronger evidence separation before termination. This sequential accumulation and boundary-based stopping distinguish DDM from decision rules based on a single observation.

We adopt these two principles: noisy evidence accumulation and boundary-based stopping, rather than the original binary formulation. DDM-VLM extends them to multiple valid joint defect states, as detailed in the following sections.

2.2. Joint-State Formulation and Visual-Memory Retrieval

To instantiate the above DDM principles for mixed-defect recognition, we formulate the task over valid joint defect

states and use retrieved wafer examples as round-specific visual memory. The overall framework is illustrated in Fig. 2.

Let $x \in \{0, 1, 2\}^{H \times W}$ denote a wafer map of size $H \times W$ and $\mathbf{y} \in \{0, 1\}^D$ its multi-hot label, where $D = 8$. We classify complete joint states, with the valid state space

$$\mathcal{Y} = \{\mathbf{y} : \mathbf{y} \text{ appears in the database}\}, \quad |\mathcal{Y}| = 38. \quad (2)$$

Before retrieval, connected defect components smaller than 10 pixels are removed, defective dies are rendered in red, and the image is resized to 384×384 . CLIP ViT-B/32 [19] extracts ℓ_2 -normalized features and FAISS [20] performs cosine nearest-neighbor search. For target image x , the top- K retrieved visual memory is

$$\mathcal{R}_K(x) = \{(x_j, \mathbf{y}_j, s_j)\}_{j=1}^K, \quad (3)$$

where s_j denotes target-example similarity. Its unique labels define the candidate space

$$\mathcal{Y}_x = \{\mathbf{y}_j : (x_j, \mathbf{y}_j, s_j) \in \mathcal{R}_K(x)\} \subseteq \mathcal{Y}. \quad (4)$$

In the main setting, $K = 10$ and the retrieved examples are divided into three disjoint index sets $\mathcal{I}_t$ with ranks 1–4, 5–7, and 8–10. At round t ,

$$\mathcal{M}_t = \{(x_j, \mathbf{y}_j, s_j) : j \in \mathcal{I}_t\}. \quad (5)$$

Thus, successive rounds introduce new visual observations. $\mathcal{Y}_x$ constrains the VLM options, while the external accumulator remains defined over the complete state space $\mathcal{Y}$.

2.3. Sequential Observation and Evidence Extraction

At round t , the VLM receives the target wafer, retrieved examples $\mathcal{M}_t$, defect descriptions, and candidate labels $\mathcal{Y}_x$. Let $p_t(\mathbf{y})$ denote the normalized answer-token probability for candidate $\mathbf{y}$. Its perceptual evidence is

$$e_t^{\text{vlm}}(\mathbf{y}) = \begin{cases} \left[\log \frac{p_t(\mathbf{y}) + \epsilon}{1/|\mathcal{Y}_x|} \right]_{-L}^L, & \mathbf{y} \in \mathcal{Y}_x, \\ 0, & \text{otherwise,} \end{cases} \quad (6)$$

where $\epsilon > 0$ is a smoothing constant and $[\cdot]_{-L}^L$ denotes clipping to $[-L, L]$.

The retrieved examples provide complementary memory evidence. For $j \in \mathcal{I}_t$, with temperature $\tau > 0$, we define

$$w_j = \exp \left(\frac{s_j - \max_{k \in \mathcal{I}_t} s_k}{\tau} \right), \quad (7)$$

and the induced label distribution

$$q_t(\mathbf{y}) = \frac{\epsilon + \sum_{j \in \mathcal{I}_t} w_j \mathbb{I}[\mathbf{y}_j = \mathbf{y}]}{\sum_{\mathbf{y}' \in \mathcal{Y}} \left(\epsilon + \sum_{j \in \mathcal{I}_t} w_j \mathbb{I}[\mathbf{y}_j = \mathbf{y}'] \right)}, \quad (8)$$

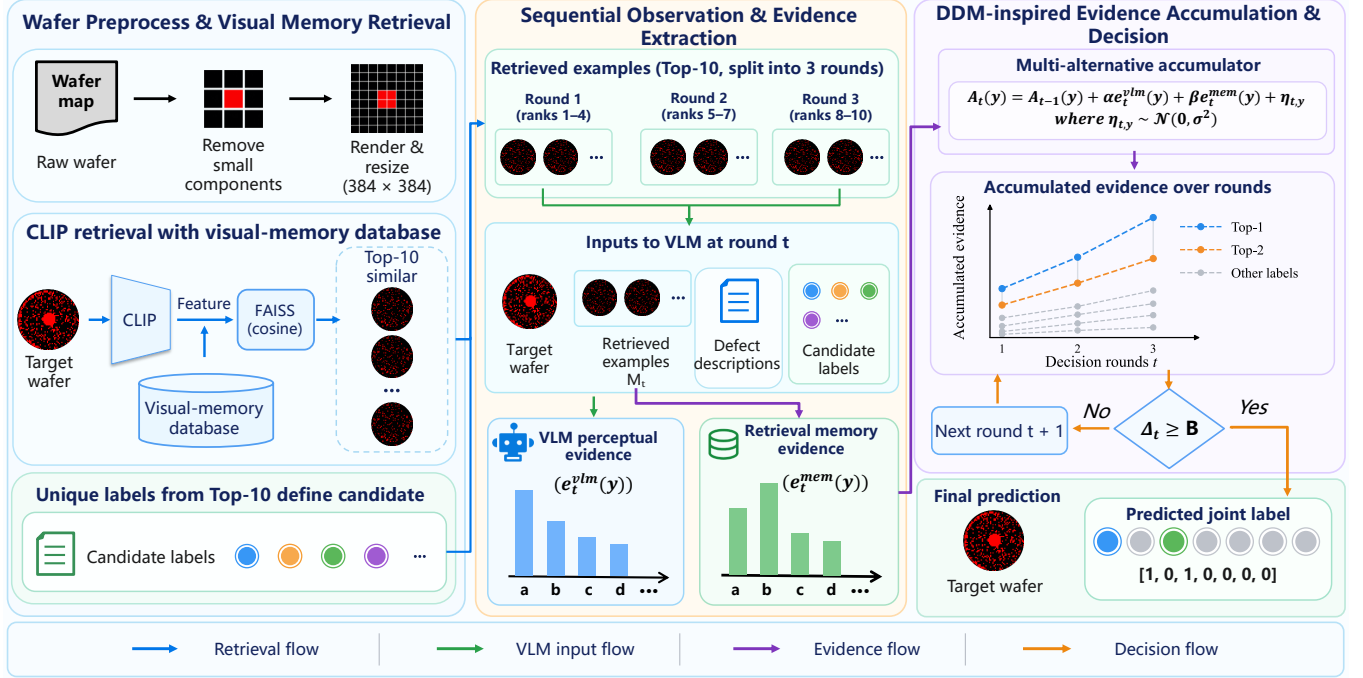


Fig. 2. Overview of DDM-VLM. CLIP-FAISS retrieval provides candidate joint states and round-specific visual memory. Perceptual and memory evidence are sequentially accumulated until the top-two evidence margin reaches the decision boundary.

where $\mathbb{I}[\cdot]$ denotes the indicator function. The corresponding memory evidence is

$$e_t^{\text{mem}}(\mathbf{y}) = \left[\log \frac{q_t(\mathbf{y})}{1/|\mathcal{Y}|} \right]_{-L}^L. \quad (9)$$

Perceptual evidence captures the VLM judgment, whereas memory evidence reflects support from visually similar labeled wafers.

2.4. DDM-Inspired Evidence Accumulation and Decision

An accumulator $A_t(\mathbf{y})$ is maintained for each $\mathbf{y} \in \mathcal{Y}$ with $A_0(\mathbf{y}) = 0$. At each round,

$$A_t(\mathbf{y}) = A_{t-1}(\mathbf{y}) + \alpha e_t^{\text{plm}}(\mathbf{y}) + \beta e_t^{\text{mem}}(\mathbf{y}) + \eta_{t,\mathbf{y}}, \quad (10)$$

where α and β weight perceptual and memory evidence, respectively, and $\eta_{t,\mathbf{y}} \sim \mathcal{N}(0, \sigma^2)$ denotes independent diffusion noise.

Let $\mathbf{y}_t^{(1)}$ and $\mathbf{y}_t^{(2)}$ be the two highest-scoring states after round t . Their evidence margin is

$$\Delta_t = A_t(\mathbf{y}_t^{(1)}) - A_t(\mathbf{y}_t^{(2)}). \quad (11)$$

Inference terminates when $\Delta_t \geq B$ or the maximum round T is reached. At the stopping round t^* , the final prediction is

$$\hat{\mathbf{y}} = \arg \max_{\mathbf{y} \in \mathcal{Y}} A_{t^*}(\mathbf{y}). \quad (12)$$

Thus, confident samples terminate early, whereas ambiguous samples receive additional observations.

DDM-VLM retains the evidence-accumulation principle of classical DDM while extending it to multiple joint states. The unchanged VLM serves as a perceptual observer, while retrieval provides complementary memory cues. The candidate set $\mathcal{Y}_x$ restricts VLM alternatives, whereas the accumulator remains defined over $\mathcal{Y}$; thus, retrieval conditions rather than determines the final decision. The boundary B controls stopping confidence, while T bounds the inference cost.

3. EXPERIMENTS

3.1. Experimental Setup

We evaluate on MixedWM38 [5], containing 38,015 wafer maps with eight-dimensional multi-hot labels and 38 observed joint states. Our main setting uses an 8:2 visual-memory-database/evaluation split. We additionally evaluate 9:1 and 7:3 splits to verify that the improvement introduced by DDM-VLM over the Pure RAG baseline remains consistent across different data partitions. Each wafer is represented as a 52×52 grid.

We evaluate Qwen2.5-VL-7B-Instruct, mPLUG-Owl3-7B-241101, Llama-3.2-11B-Vision-Instruct, and InternVL3-8B. The proposed framework uses the top- $K = 10$ CLIP ViT-B/32 retrieved wafers as visual memory and divides them into (4, 3, 3) successive observation sets. We set $\alpha = 0.8$,

Table 1. Main results and additional analyses (EM, %).
(a) Main results under the 8:2 split.

Method	Components		EM (%)			
	Memory	DDM	Qwen	mPLUG	Llama	InternVL
Direct VLM	×	×	2.66	2.66	1.33	4.84
Pure RAG	✓	×	52.92	50.13	41.52	44.69
Ours	✓	✓	54.53	53.78	50.63	51.35

(b) Boundary sensitivity

B	EM (%)	Avg. calls
1.0	53.80	1.60
1.5	54.53	2.00
2.0	54.09	2.43
2.5	53.93	2.67

(c) Evidence ablation

Variant	EM (%)
Ours	54.53
w/o e_t^{mem}	54.19
w/o e_t^{vlm}	41.98

(d) Effectiveness across data splits.

Split	Method	Qwen	mPLUG	Llama	InternVL
9:1	Pure RAG	54.97	51.92	43.71	45.26
	Ours	55.89	56.50	52.68	53.63
8:2	Pure RAG	52.92	50.13	41.52	44.69
	Ours	54.53	53.78	50.63	51.35
7:3	Pure RAG	54.06	50.71	41.80	46.90
	Ours	55.35	56.38	51.42	52.80

$\beta = 0.2$, $B = 1.5$, $\sigma = 0.05$, $\tau = 0.05$, $L = 3.0$, and $T = 3$. Performance is measured by exact-match accuracy (EM), which requires the complete joint defect label to be predicted correctly. We also report the average number of VLM calls when analyzing the decision boundary.

3.2. Baseline Methods

Direct VLM predicts the joint label from only the target wafer and defect definitions. **Pure RAG** uses the same CLIP-FAISS retrieval and top- K visual memory as ours, but presents all examples in one VLM interaction and directly adopts its response. It therefore controls for retrieval while removing sequential evidence accumulation.

3.3. Main Results

Table 1(a) compares three decision paradigms under the main 8:2 split. Direct VLM performs poorly on the unconstrained joint-state task, highlighting the difficulty of making a reliable mixed-defect decision from a single VLM response. More importantly, under the same retrieval condition, the proposed DDM mechanism consistently improves all four VLM backbones. EM increases from 52.92% to 54.53% for Qwen2.5-VL-7B-Instruct and from 41.52% to 50.63% for Llama-3.2-11B-Vision, indicating that the gain cannot be attributed to retrieval alone and supporting the benefit of sequential evidence accumulation. Fig. 3 provides a representative example,

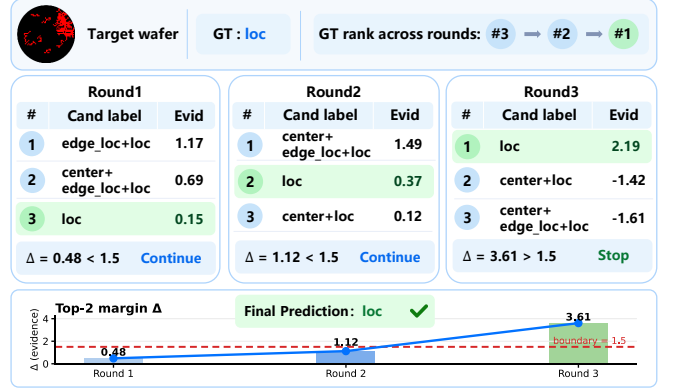


Fig. 3. Three-round decision trajectory. The ground-truth state *loc* rises from rank 3 to 1, while the top-two margin increases from 0.48 to 3.61 and crosses $B = 1.5$ at Round 3.

where the ground-truth state *loc* progressively rises from rank 3 to rank 1 across three rounds.

Using Qwen2.5-VL-7B-Instruct, Table 1(b) examines the sensitivity to the decision boundary B . Among the tested values, $B = 1.5$ achieves the highest EM of 54.53% and is therefore adopted as the default setting. We additionally report the average number of VLM calls to characterize the stopping behavior. As B increases from 1.0 to 2.5, the average number of calls rises from 1.60 to 2.67. This is consistent with the proposed boundary-based mechanism: a smaller B allows more samples to terminate once a modest top-two evidence margin is established, whereas a larger B requires stronger separation and consequently causes more samples to proceed to later rounds.

On the same backbone, Table 1(c) evaluates the two evidence sources. Removing perceptual evidence by setting $\alpha = 0$ reduces EM from 54.53% to 41.98%, confirming that VLM perception provides the primary decision signal. Removing memory evidence by setting $\beta = 0$ results in a smaller decrease to 54.19%, showing that visual memory provides complementary support to the sequential decision process.

Finally, Table 1(d) shows that DDM-VLM consistently outperforms the corresponding Pure RAG baseline under different data partitions, indicating that the benefit of sequential evidence accumulation is not specific to the main 8:2 split.

4. CONCLUSION

We presented DDM-VLM, a psychology-inspired sequential decision framework for joint wafer defect recognition. DDM-VLM combines multi-round evidence accumulation with boundary-based stopping and consistently outperforms retrieval-augmented single-response baselines across multiple VLM backbones on MixedWM38, supporting sequential decision making for mixed-defect recognition.

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